

MARRAKECH MENARA, Morocco

WMO: 602300

Lat: 31.607N

Lon: 8.036W

Elev: 1535

StdP: 13.90

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.2	41.4	20.3	16.1	72.3	24.9	20.0	69.2	18.7	57.0	15.0	56.6	2.8	80	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	28.0	107.8	67.9	104.0	67.9	100.4	67.8	72.8	95.4	71.5	93.5	70.4	91.4	9.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	103.3	78.1	65.3	98.8	77.3	64.2	95.3	76.8	37.3	96.0	36.1	93.4	35.1	91.8	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.2	12.1	DB	35.1	112.6	3.3	2.8	32.7	114.7	30.8	116.3	29.0	117.9	26.6	119.9	(4)
(5)			WB	32.9	75.3	3.2	3.0	30.6	77.5	28.8	79.2	27.0	80.9	24.7	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.4	54.8	57.6	62.9	66.4	72.3	78.9	84.4	84.7	77.9	72.5	62.8	56.7	(6)
(7)		DBStd	11.66	4.34	4.84	5.88	6.11	7.09	6.28	6.16	5.65	5.20	5.60	5.72	4.02	(7)
(8)		HDD50	17	10	4	0	0	0	0	0	0	0	0	0	3	(8)
(9)		HDD65	1080	318	209	110	53	12	0	0	0	0	8	110	260	(9)
(10)		CDD50	7087	157	217	401	491	690	866	1066	1075	836	696	383	209	(10)
(11)		CDD65	2675	0	2	46	94	237	416	601	610	386	239	43	1	(11)
(12)		CDH74	32351	25	108	658	1132	2683	4951	8081	8034	4004	2214	430	31	(12)
(13)		CDH80	17971	2	10	181	456	1373	2801	5114	4966	2027	943	97	1	(13)
(14)	Wind	WSAvg	5.3	4.0	4.8	5.7	6.1	6.2	6.5	6.1	6.0	5.5	4.9	4.3	3.8	(14)
(15)	Precipitation	PrecAvg	8.9	1.0	1.1	1.5	1.0	0.4	0.1	0.1	0.2	0.6	0.8	1.2	1.0	(15)
(16)		PrecMax	17.0	4.9	4.2	5.7	4.6	2.3	1.1	1.3	1.7	3.1	2.7	5.7	3.0	(16)
(17)		PrecMin	4.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.2	1.1	1.1	1.2	1.2	0.5	0.2	0.2	0.4	1.0	0.7	1.2	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	80.9	89.9	95.4	104.2	109.1	112.7	111.4	104.2	96.9	86.4	77.4	(19)
(20)			MCWB	54.6	55.8	59.0	60.6	64.6	66.3	68.6	67.9	66.9	64.2	60.3	56.5	(20)
(21)		2%	DB	73.1	77.3	84.5	90.0	98.5	103.2	108.9	107.2	98.8	93.0	82.7	73.6	(21)
(22)			MCWB	53.4	55.0	58.3	60.7	64.4	66.6	68.6	68.4	66.9	63.3	58.9	55.3	(22)
(23)		5%	DB	69.8	73.8	80.8	85.6	93.0	98.4	104.2	102.7	94.9	89.4	79.1	70.6	(23)
(24)			MCWB	53.0	55.1	57.2	60.1	63.6	66.8	68.5	69.0	66.6	63.1	58.5	55.0	(24)
(25)		10%	DB	66.4	70.2	77.2	80.6	87.8	93.6	100.0	98.9	91.0	85.6	75.2	67.8	(25)
(26)			MCWB	52.8	54.0	56.9	59.7	63.0	66.4	68.5	68.6	66.2	62.9	58.1	54.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.4	61.7	63.1	66.2	68.9	72.6	76.0	74.6	71.8	69.9	67.2	61.5	(27)
(28)			MCD B	68.0	70.0	76.7	84.6	88.8	95.6	99.2	96.2	91.2	84.2	74.7	67.8	(28)
(29)		2%	WB	57.4	59.1	61.3	63.8	67.1	70.4	72.8	72.9	70.4	67.8	63.5	59.1	(29)
(30)			MCD B	66.6	69.1	75.6	81.5	87.6	92.3	97.1	94.7	87.7	81.4	73.9	67.4	(30)
(31)		5%	WB	55.6	57.3	59.9	62.1	65.6	69.0	71.2	71.7	69.1	66.4	61.4	57.4	(31)
(32)			MCD B	64.8	68.6	74.3	78.8	85.9	90.4	94.5	93.7	85.5	79.9	72.5	65.8	(32)
(33)		10%	WB	53.8	55.6	58.5	60.6	64.2	67.6	70.0	70.6	68.0	65.1	59.8	55.8	(33)
(34)			MCD B	62.9	66.7	72.2	76.4	83.7	88.6	92.8	91.9	83.9	78.2	71.0	64.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.1	22.3	23.1	22.7	24.5	27.2	29.6	23.7	22.1	21.2	21.9	(35)	
(36)			MCDBR	28.3	29.2	29.8	30.1	31.7	31.6	32.3	31.0	28.9	28.0	26.8	26.9	(36)
(37)			MCWBR	13.4	12.6	10.7	9.3	9.3	8.5	8.2	7.8	7.6	8.9	10.5	12.7	(37)
(38)		5% WB	MCD BR	22.2	23.8	24.6	26.3	28.2	29.0	30.4	28.5	24.8	22.8	20.8	20.7	(38)
(39)			MCWBR	11.9	11.5	10.1	9.4	9.2	8.9	8.8	8.1	7.5	8.2	9.2	11.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.348	0.413	0.411	0.447	0.488	0.558	0.539	0.453	0.410	0.347	0.312	(40)	
(41)		taud	2.508	2.361	2.134	2.183	2.089	1.984	1.806	1.871	2.106	2.226	2.421	2.513	(41)	
(42)		Ebn at Noon	294	288	274	279	269	257	238	241	259	262	274	285	(42)	
(43)		Edh at Noon	28	35	47	47	52	57	68	63	48	40	30	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1141	1412	1783	2094	2302	2495	2437	2198	1881	1480	1178	1041	(44)	
(45)		RadStd	76	109	92	128	153	115	79	73	96	69	81	67	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.83	N/A	-2.48	N/A	N/A	N/A	N/A	+275	+308
(47) Regional (6 neighbors)		+0.58	N/A	-2.14	+1.28	-0.68	N/A	N/A	+275	+308

Nomenclature: See separate page